

SHELBY-CLEVELAND COUNTY AP, NC, USA

WMO: 720277

Lat: 35.256N

Lon: 81.601W

Elev: 847

StdP: 14.25

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63843

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.4	23.7	1.6	6.1	28.5	6.1	7.8	31.2	22.8	44.6	19.3	45.9	3.4	30	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	93.3	73.3	90.9	72.6	89.4	72.2	76.7	86.0	75.8	85.0	74.9	83.8	5.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.1	131.4	78.9	73.2	127.2	78.4	72.5	124.2	78.0	40.6	85.5	39.8	85.4	38.9	83.8	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.7	15.0	12.2	DB	12.9	96.3	6.6	3.3	8.2	98.7	4.3	100.6	0.6	102.5	-4.2	104.9
			WB	11.4	78.1	6.0	1.3	7.1	79.1	3.6	79.9	0.3	80.6	-4.1	81.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	40.5	44.7	51.4	60.4	68.3	76.1	78.2	77.1	71.9	60.8	49.8	44.6	(6)
	DBStd	15.12	9.52	9.25	9.07	7.31	6.00	4.20	3.59	4.30	5.32	7.78	7.81	8.86	(7)
	HDD50	945	320	196	95	10	0	0	0	0	0	10	96	218	(8)
	HDD65	3241	760	569	427	171	40	0	0	0	10	175	457	632	(9)
	CDD50	4743	24	49	139	323	568	783	875	839	658	343	91	51	(10)
	CDD65	1567	0	1	7	34	143	333	410	374	218	44	2	1	(11)
	CDH74	13764	0	7	78	402	1206	3044	3728	3215	1705	356	17	6	(12)
	CDH80	5168	0	0	6	65	311	1234	1598	1301	581	71	1	0	(13)
(14) Wind	WSAvg	4.3	4.9	5.3	5.6	5.5	4.4	3.6	3.3	3.0	3.8	4.0	4.0	4.1	(14)
(15) Precipitation	PrecAvg	48.3	4.3	3.5	5.0	4.1	3.5	4.5	4.2	4.5	3.9	3.6	3.3	4.1	(15)
	PrecMax	68.3	7.5	7.5	8.7	10.1	8.6	10.6	10.2	7.8	11.7	12.1	8.3	8.0	(16)
	PrecMin	27.9	1.0	0.9	1.3	1.1	0.9	0.9	1.4	0.6	0.1	0.0	1.1	1.9	(17)
	PrecStd	9.9	2.0	1.5	1.9	2.2	1.9	2.5	1.9	2.2	2.4	2.9	1.8	1.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.8	74.5	80.9	85.7	90.1	95.5	95.9	97.3	91.5	87.2	76.8	72.9	(19)
		MCWB	59.9	61.8	60.6	63.4	69.0	71.9	73.2	71.7	70.6	70.8	62.5	61.3	(20)
	2%	DB	64.0	69.3	76.1	81.6	86.3	91.9	92.7	93.0	89.6	81.3	72.0	67.6	(21)
		MCWB	56.5	58.0	59.2	62.5	67.8	72.2	73.9	73.7	70.4	66.6	59.0	60.5	(22)
	5%	DB	60.6	64.5	72.0	79.0	83.4	90.0	90.6	90.1	86.3	78.1	68.1	63.5	(23)
		MCWB	53.1	55.2	57.2	61.8	68.1	71.9	73.6	73.3	70.5	65.6	58.8	57.5	(24)
	10%	DB	55.4	61.2	67.8	75.4	81.0	87.2	88.4	87.1	83.1	74.2	64.1	59.4	(25)
		MCWB	49.6	52.2	55.3	60.4	66.4	71.4	73.2	73.2	70.2	64.7	56.5	53.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.7	64.1	64.1	69.3	73.4	76.9	77.7	77.8	75.4	74.1	68.1	66.5	(27)
		MCDB	65.5	70.9	72.4	75.3	81.2	85.8	87.6	86.9	84.0	79.5	70.3	68.6	(28)
	2%	WB	59.3	61.4	62.2	66.9	71.5	75.3	76.8	76.7	74.3	71.5	64.1	62.9	(29)
		MCDB	63.2	66.0	69.3	75.2	78.9	85.3	86.7	85.5	82.4	76.5	67.1	66.0	(30)
	5%	WB	55.2	58.3	60.2	64.8	70.4	74.4	75.9	75.7	73.1	69.1	61.5	59.4	(31)
		MCDB	58.5	63.3	68.0	73.4	78.1	84.2	85.5	84.4	80.9	73.7	65.2	62.0	(32)
	10%	WB	50.9	53.8	57.9	63.1	69.1	73.4	75.0	74.7	71.9	66.9	58.4	54.7	(33)
		MCDB	54.8	58.6	66.1	71.2	76.9	82.8	84.1	82.9	79.2	72.7	63.6	58.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	19.9	21.3	22.6	20.8	20.1	19.1	18.5	19.0	21.3	21.9	18.2	(35)
		MCDBR	24.0	25.2	25.8	27.2	24.1	23.4	22.1	22.4	23.8	24.4	26.3	22.0	(36)
		MCWBR	16.5	15.7	12.8	11.6	9.2	7.4	6.5	6.4	8.1	11.5	15.3	14.6	(37)
	5% WB	MCDBR	18.9	21.4	21.1	21.5	18.5	19.8	18.8	18.4	18.7	18.1	18.8	16.5	(38)
		MCWBR	16.0	15.8	13.3	11.1	8.4	7.5	6.6	6.4	7.7	10.0	14.5	13.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.309	0.317	0.352	0.399	0.454	0.482	0.528	0.516	0.421	0.373	0.328	0.310	(40)
	taud		2.511	2.480	2.418	2.281	2.162	2.122	2.020	2.034	2.286	2.403	2.490	2.536	(41)
	Ebn at Noon		285	295	291	281	266	257	245	245	265	270	275	277	(42)
	Edh at Noon		26	30	35	42	48	50	55	53	39	32	26	24	(43)
(44) All-Sky Solar Radiation	RadAvg		836	1042	1375	1711	1899	2011	1877	1732	1477	1222	925	717	(44)
	RadStd		48	116	96	132	154	144	128	84	134	146	78	69	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (53 neighbors)	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+301	+194

Nomenclature: See separate page